



Lesson 3.4 — Graphing Polynomials

Big idea: To sketch a polynomial: find zeros (with multiplicities), find the y-intercept, and use end behavior to connect the dots. The smooth curve has at most (degree – 1) turning points.

Quick review

Step 1: degree + leading coefficient → end behavior.

Step 2: factor and find zeros + multiplicities → cross/bounce behavior on the x-axis.

Step 3: y-intercept = $P(0)$.

Step 4: sign chart: between zeros the polynomial doesn't change sign (unless it has a zero of even multiplicity, where it doesn't change sign through that zero).

Turning points: a polynomial of degree n has at most $n - 1$ turning points (local max/min).

Worked example

Worked example. Sketch $P(x) = (x + 2)(x - 1)^2$.

Degree 3, leading + → down/up. Zeros: $x = -2$ (mult 1, crosses), $x = 1$ (mult 2, bounces). y-intercept: $(2)(-1)^2 = 2$.

Graph comes from below, crosses at $x = -2$, rises, touches $x = 1$ from above and bounces, continues up.

Warm-ups

1. How many turning points can a degree-5 polynomial have at most?
2. State the y-intercept of $P(x) = (x - 3)(x + 1)(x - 5)$.
3. Find the zeros of $P(x) = x^3 - 4x$.

Practice

1. Sketch (describe) $P(x) = x(x - 3)(x + 2)$. Include zeros, end behavior, and y-intercept.
2. Sketch (describe) $P(x) = -(x + 1)^2(x - 2)$.
3. Find zeros, multiplicities, and y-intercept of $P(x) = x^4 - 5x^2 + 4$.

Challenge

1. A polynomial has degree 4, leading coefficient 1, zeros at -1 , 2 (multiplicity 2), and 4, with y-intercept 16. Write the equation. (Hint: check sign of constant.)
2. Sketch (describe) $P(x) = x^3(x - 2)^2(x + 3)$. Include all behaviors.



Answer key — Lesson 3.4

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. At most 4 turning points.
2. $(-3)(1)(-5) = 15$.
3. $x(x^2 - 4) = x(x - 2)(x + 2) \Rightarrow x = 0, 2, -2$.

Practice

1. Degree 3, leading $+$ $\rightarrow$ down/up. Zeros 0, 3, -2 (each mult 1, all crosses). y-intercept (0). Order on x-axis: $-2, 0, 3$. Goes from $-\infty$ up through -2 , peak, down through 0, valley, up through 3 to $+\infty$.
2. Degree 3, leading $-$ (because of the $-$) $\rightarrow$ up/down. Zeros: $x = -1$ (mult 2, bounces), $x = 2$ (mult 1, crosses). y-intercept: $-(1)(-2) = 2$. Comes from $+\infty$, drops to touch $x = -1$ and bounces down, continues to cross at $x = 2$ and goes to $-\infty$.
3. Factor: $(x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$. Zeros: $\pm 1, \pm 2$ (all simple). y-intercept = 4.

Challenge

1. $P(x) = (x + 1)(x - 2)^2(x - 4)$. Check $P(0) = (1)(4)(-4) = -16$. We need $+16$, so add a leading factor -1 : $P(x) = -(x + 1)(x - 2)^2(x - 4)$. Check: $-(1)(4)(-4) = 16$ ✓. (But leading coefficient becomes -1 , conflicting with the "1" given. The problem is not consistent — flag this. To get $y = 16$ AND leading 1, the y-intercept must be $(1)(-2)^2(-4) = -16$, contradiction.) Any consistent choice with leading coefficient ± 1 must accept the y-intercept that comes out: y-intercept = ± 16 ; with $+16$ you need leading -1 .
2. Degree 6, leading $+$ $\rightarrow$ up/up. Zeros: 0 (mult 3, crosses, flatter), 2 (mult 2, bounces), -3 (mult 1, crosses). y-intercept = 0. Order on x-axis: $-3, 0, 2$. Comes from $+\infty$, dips through -3 , comes back up, crosses 0 with flatter 'S', climbs, touches 2 and bounces, returns to $+\infty$.